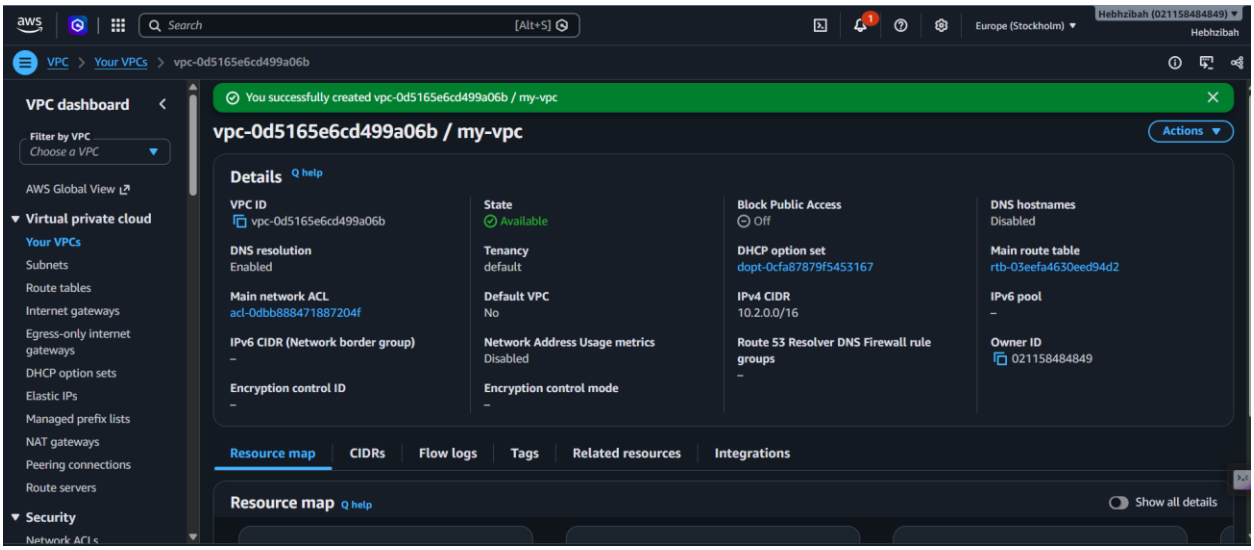
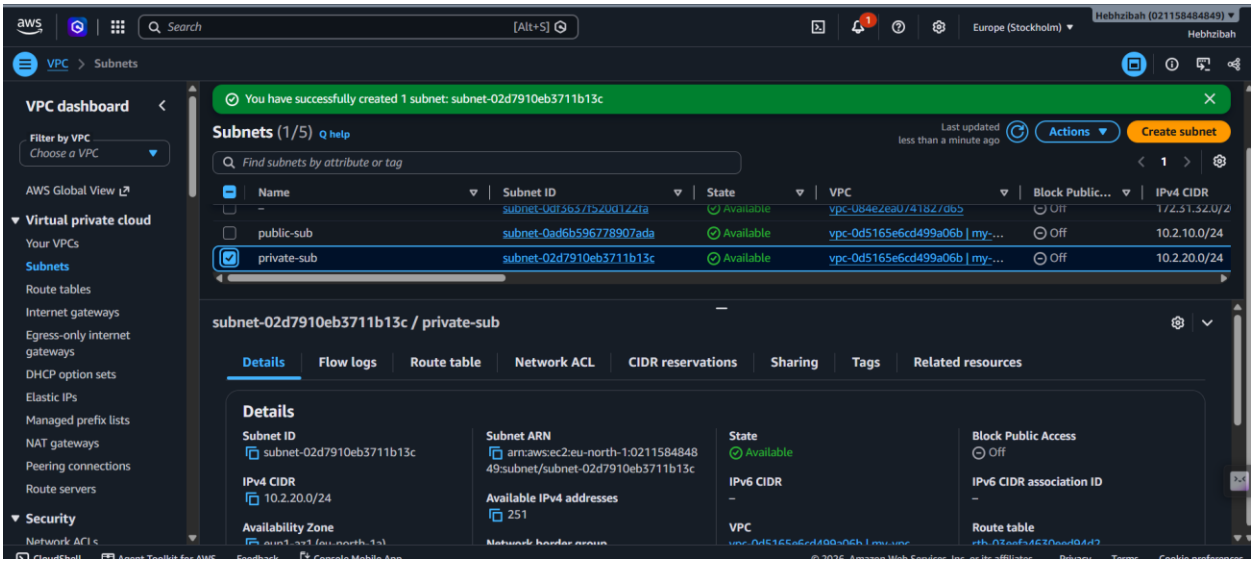
Design and implement a basic AWS Virtual Private Cloud (VPC)

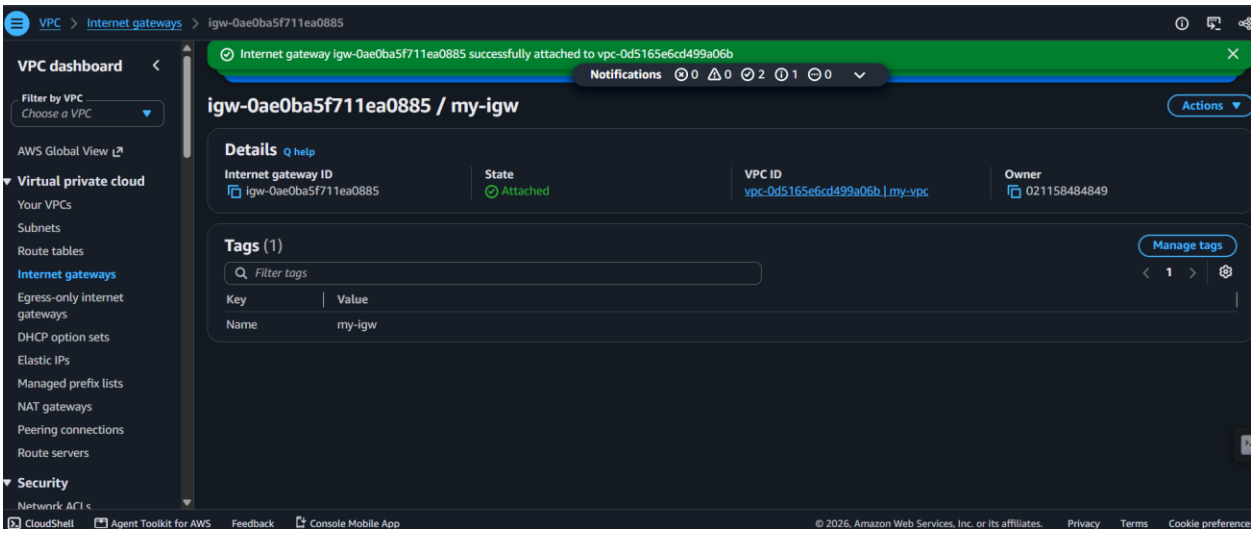
Create VPC:



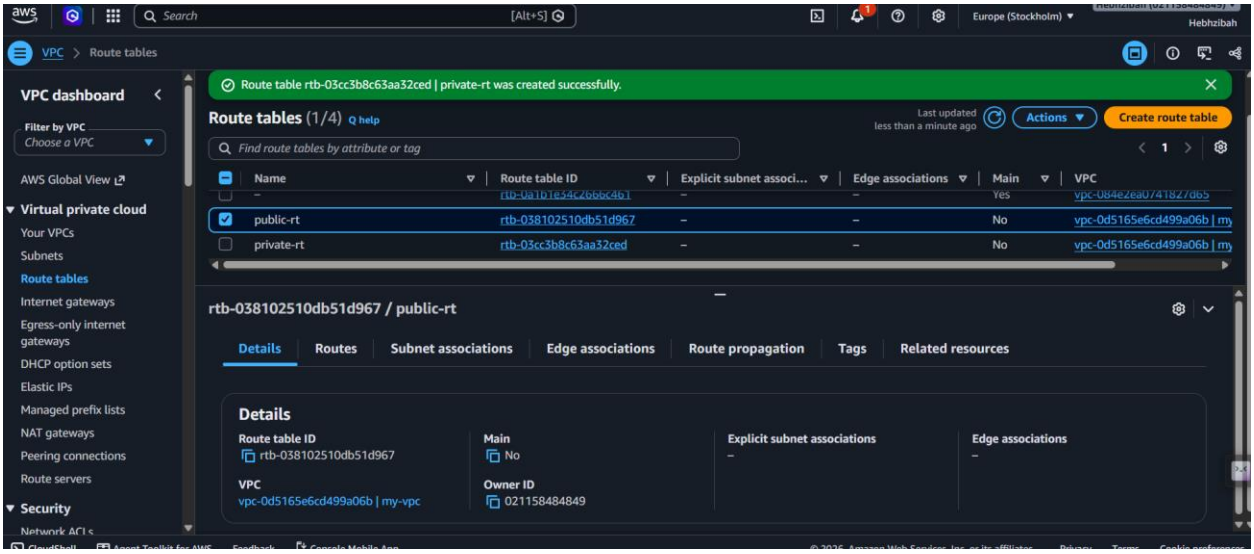
Create Subnets:



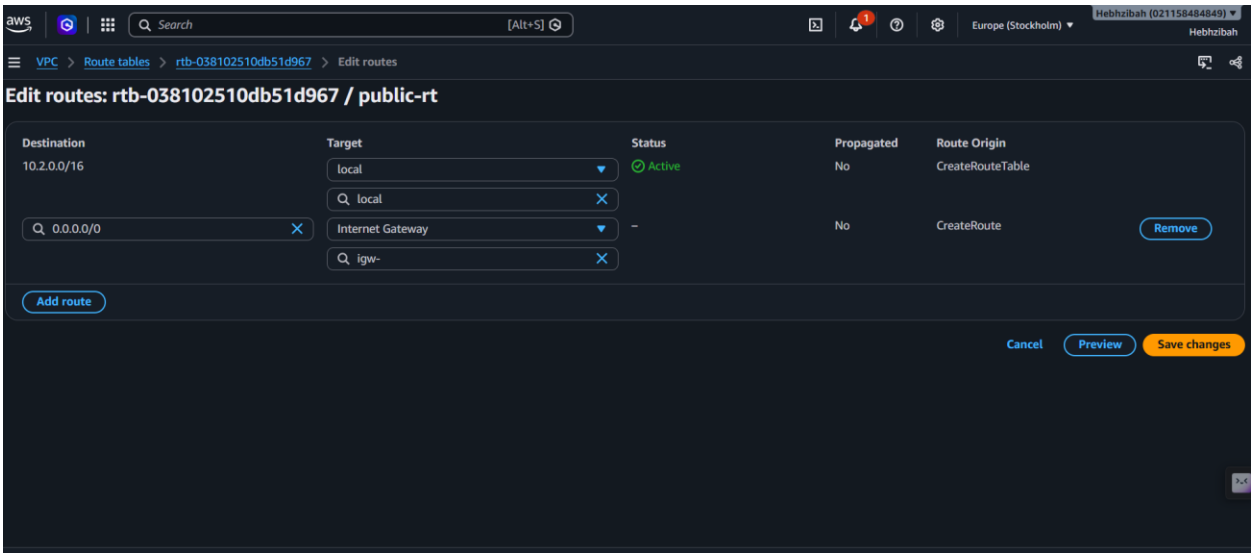
Create and attach an IGW to the VPC:



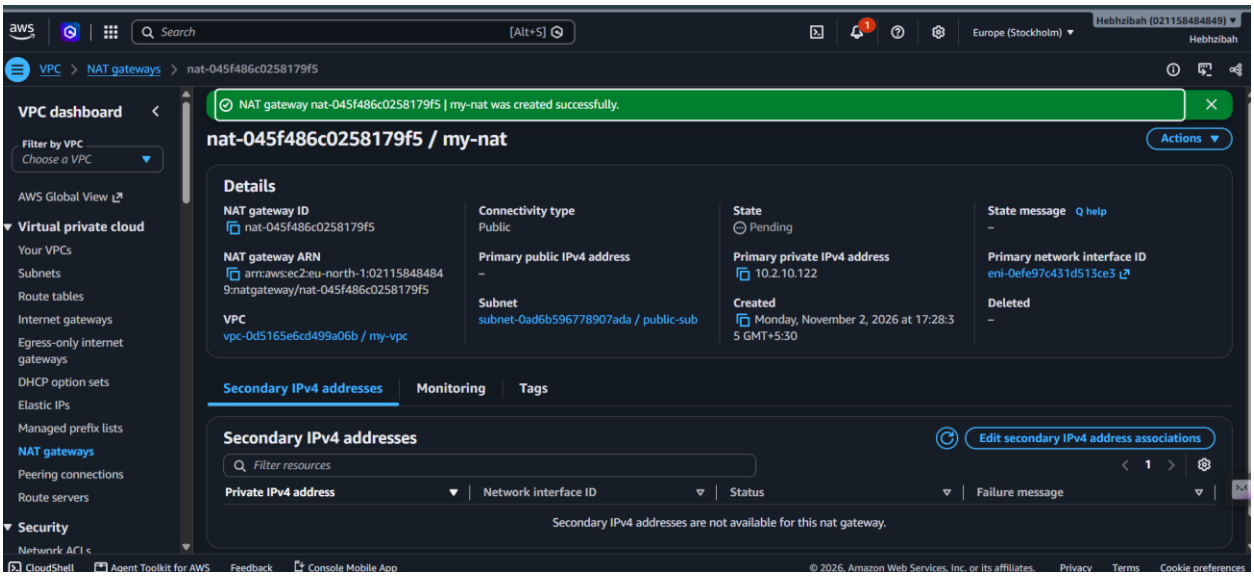
Configure Route Tables:



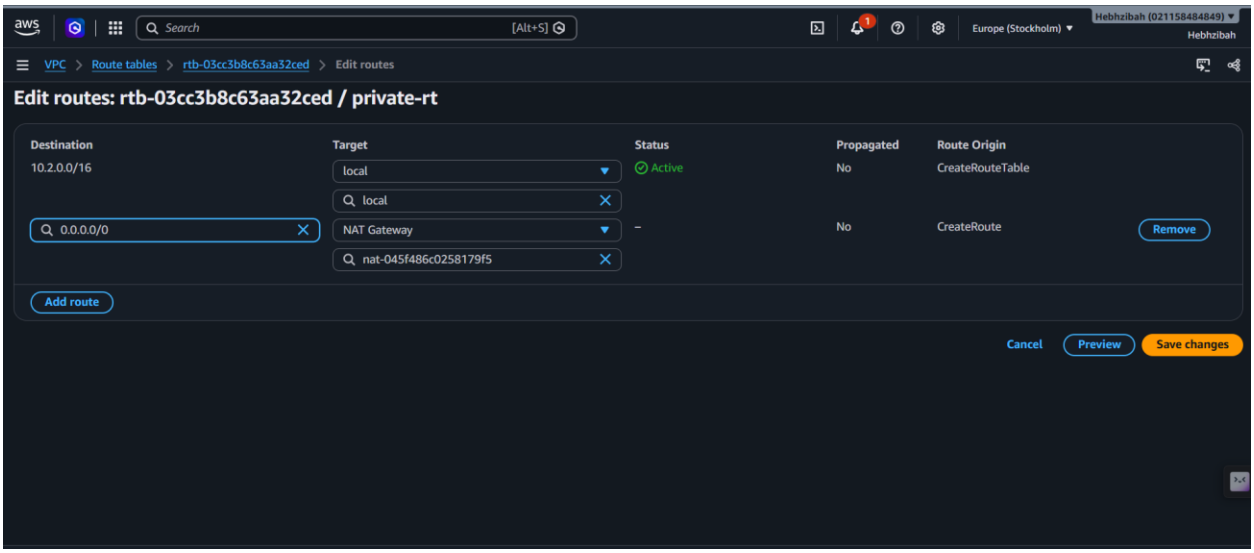
Associate with Public Subnet and add 0.0.0.0/0 route through IGW:



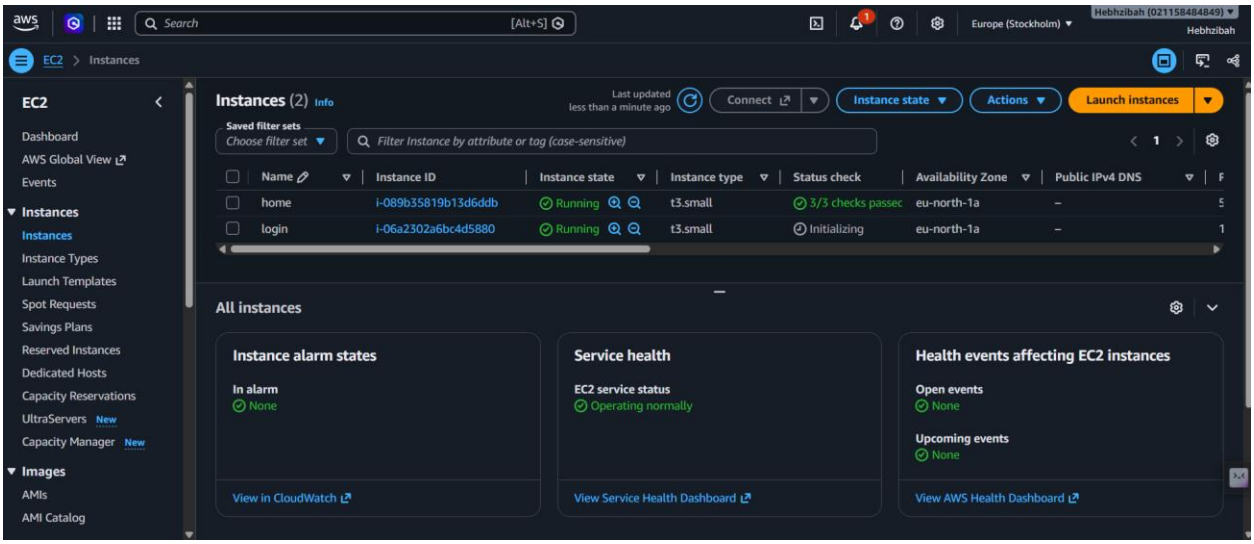
Configure NAT Gateway:



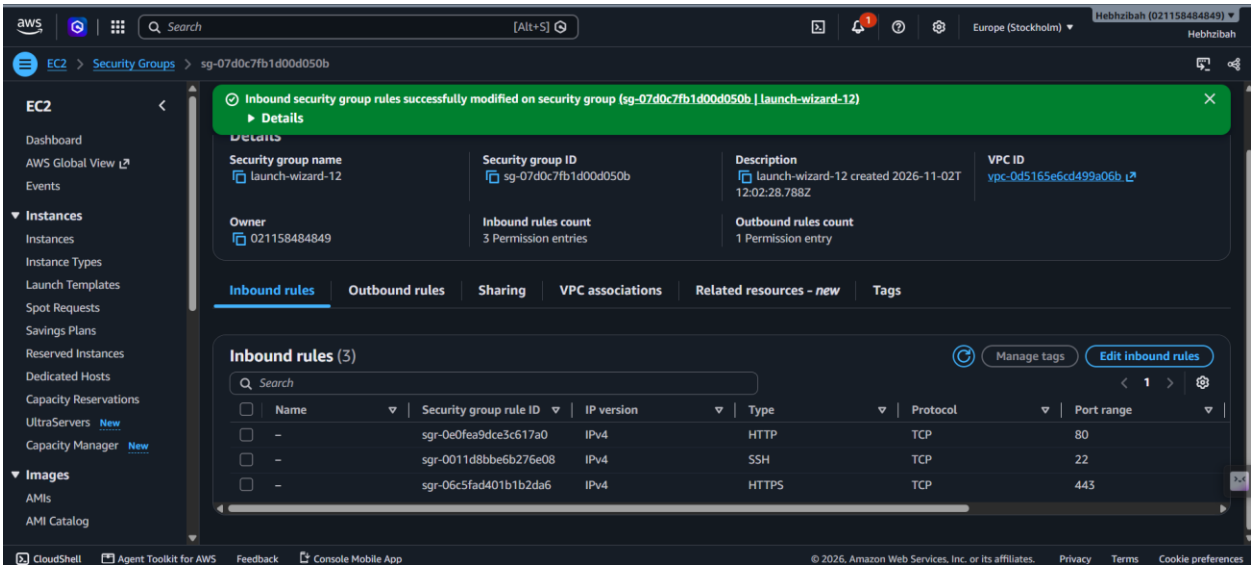
Associate with Private Subnet and configure outbound internet access through NAT Gateway:



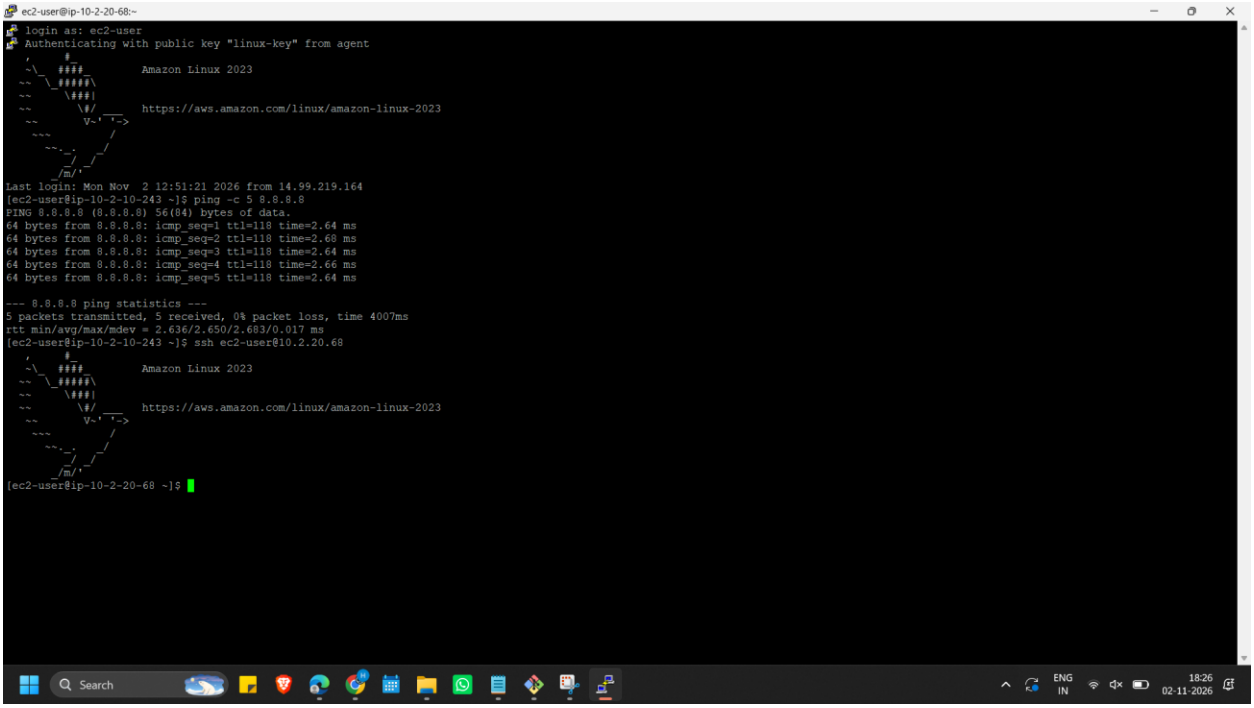
Created instances:

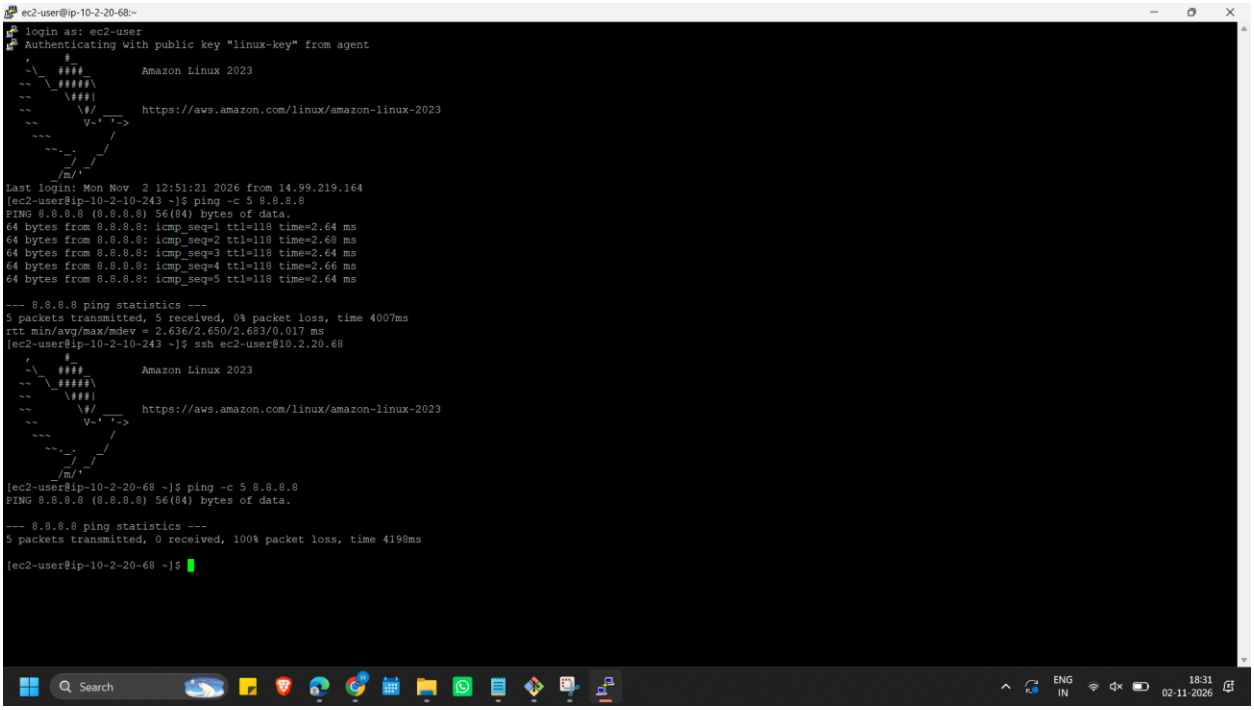


Configure Security Groups:

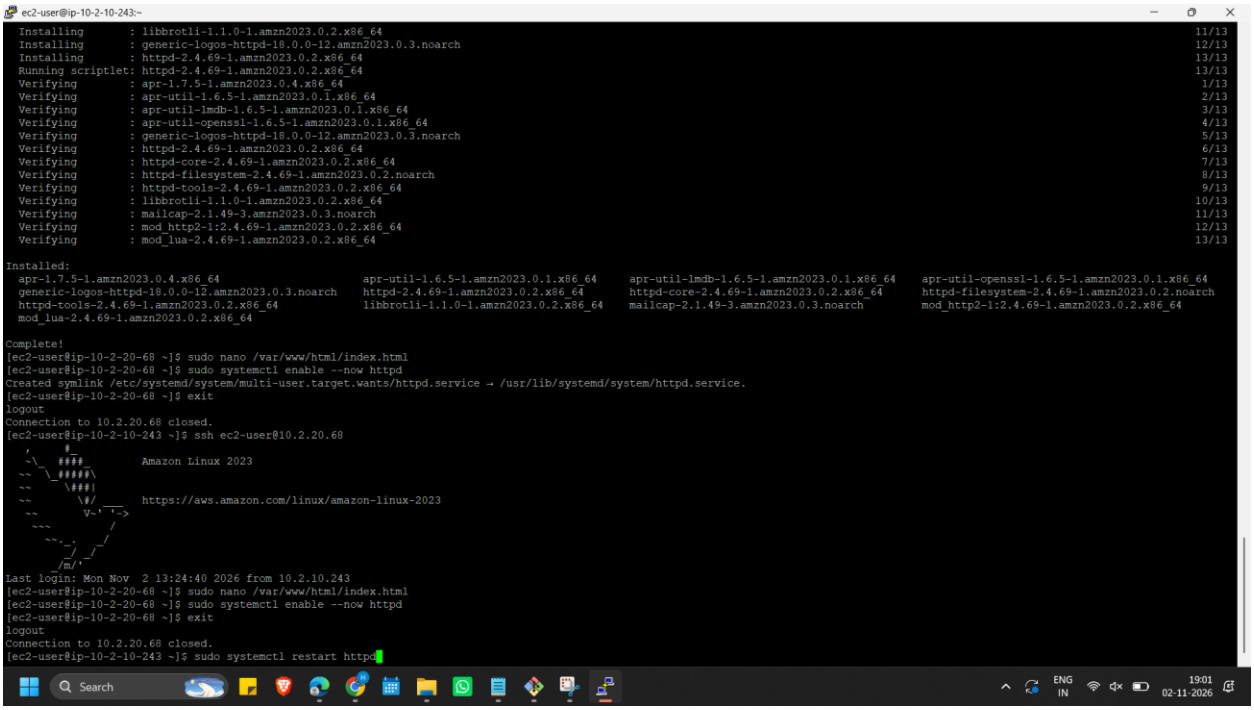


EC2 connectivity and network isolation verification:


ec2-user@ip-10-2-20-68:~
login as: ec2-user
Authenticating with public key "linux-key" from agent
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Mon Nov 2 12:51:21 2026 from 14.99.219.164
[ec2-user@ip-10-2-10-243 ~]$ ping -c 5 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=2.64 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=2.68 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=2.64 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=2.66 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=2.64 ms
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 2.636/2.650/2.683/0.017 ms
[ec2-user@ip-10-2-10-243 ~]$ ssh ec2-user@10.2.20.68
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-2-20-68 ~]$
Search
ENG
IN
18:26
02-11-2026


ec2-user@ip-10-2-20-68:~
login as: ec2-user
Authenticating with public key "linux-key" from agent
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Mon Nov 2 12:51:21 2026 from 14.99.219.164
[ec2-user@ip-10-2-10-243 ~]$ ping -c 5 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=2.64 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=2.68 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=2.64 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=2.66 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=2.64 ms
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 2.636/2.650/2.683/0.017 ms
[ec2-user@ip-10-2-10-243 ~]$ ssh ec2-user@10.2.20.68
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-2-20-68 ~]$ ping -c 5 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4198ms
[ec2-user@ip-10-2-20-68 ~]$
Search
ENG
IN
18:31
02-11-2026


Not secure
13.48.5.93
Paused
All Bookmarks
GNU nano 8.3 /var/www/html/index.html
Welcome from homepage
Go to Login Page


ec2-user@ip-10-2-10-243:~
Installing : libbrotli-1.1.0-1.amzn2023.0.2.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.69-1.amzn2023.0.2.x86_64 13/13
Running scriptlet: httpd-2.4.69-1.amzn2023.0.2.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.5-1.amzn2023.0.1.x86_64 2/13
Verifying : apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64 3/13
Verifying : apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.69-1.amzn2023.0.2.x86_64 6/13
Verifying : httpd-core-2.4.69-1.amzn2023.0.2.x86_64 7/13
Verifying : httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch 8/13
Verifying : httpd-tools-2.4.69-1.amzn2023.0.2.x86_64 9/13
Verifying : libbrotli-1.1.0-1.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64 12/13
Verifying : mod_lua-2.4.69-1.amzn2023.0.2.x86_64 13/13
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64
apr-util-1.6.5-1.amzn2023.0.1.x86_64
apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.69-1.amzn2023.0.2.x86_64
httpd-core-2.4.69-1.amzn2023.0.2.x86_64
httpd-filesystem-2.4.69-1.amzn2023.0.2.noarch
httpd-tools-2.4.69-1.amzn2023.0.2.x86_64
libbrotli-1.1.0-1.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_http2-1:2.4.69-1.amzn2023.0.2.x86_64
mod_lua-2.4.69-1.amzn2023.0.2.x86_64
Complete!
[ec2-user@ip-10-2-20-68 ~]$ sudo nano /var/www/html/index.html
[ec2-user@ip-10-2-20-68 ~]$ sudo systemctl enable --now httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-2-20-68 ~]$ exit
logout
Connection to 10.2.20.68 closed.
[ec2-user@ip-10-2-10-243 ~]$ ssh ec2-user@10.2.20.68
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Mon Nov 2 13:24:40 2026 from 10.2.10.243
[ec2-user@ip-10-2-20-68 ~]$ sudo nano /var/www/html/index.html
[ec2-user@ip-10-2-20-68 ~]$ sudo systemctl enable --now httpd
[ec2-user@ip-10-2-20-68 ~]$ exit
logout
Connection to 10.2.20.68 closed.
[ec2-user@ip-10-2-10-243 ~]$ sudo systemctl restart httpd
Search
ENG
IN
19:01
02-11-2026



hi form login page